Julie Xian

MD/PhD candidate in the Johns Hopkins Medical Scientist Training Program
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EDUCATION

Johns Hopkins Medicine, Baltimore, Maryland 2023–present
Medical Scientist Training Program, MD/PhD candidate
PhD candidate in the Department of Biomedical Engineering
Advisor: Jean Fan, PhD | Center for Computational Biology
Co-advisor: Valsamo Anagnostou, MD, PhD | Sidney Kimmel Cancer Center
Research focus: *Delineating cellular niches in the tumor microenvironment through integration of spatial transcriptomics and computational pathology*

University of Pennsylvania, Philadelphia, Pennsylvania 2018-2021
B.A. in Neuroscience | Minor in Chemistry

AWARDS & HONORS

2025 Finalist for the Paul & Daisy Soros Fellowships for New Americans
2024 American Academy of Neurology Medical Student Experience at the Annual Meeting Scholarship
2023 Rare Epilepsy Network Young Clinician Investigator
2022 University of Pennsylvania Emerging Scholar Talk Award
2021 American Epilepsy Society Grass Foundation Young Investigator Award
2021 University of Pennsylvania College of Arts and Sciences Travel Grant
2021 Summa Cum Laude, University of Pennsylvania

PUBLICATIONS

1. J.L. McKee, J.H. Magielski, **J. Xian**, ..., I. Helbig. "Clinical signatures of *SYNGAP1*-related disorders through data integration." *Genetics in Medicine*, 2025.
2. J.H. Magielski, S. Cohen, M.C. Kaufman, S. Parthasarathy, **J. Xian**, ..., J.L. McKee. "Deciphering the natural history of *SCN8A*-related disorders." *Neurology*, 2025.
3. J.H. Magielski, S.M. Ruggiero, **J. Xian**, ..., I. Helbig. "The clinical and genetic spectrum of paediatric speech and language disorders." *Brain*, 2025.
4. P.D. Galer, S. Parthasarathy, **J. Xian**, ..., I. Helbig. "Clinical signatures of genetic epilepsies precede diagnosis in electronic medical records of 32,000 individuals." *Genetics in Medicine*, 2024.
5. K.M. Thalwitzer, **J. Xian**, D. de Campo, ..., I. Helbig. "Early life seizures and epileptic spasms in *STXBP1*-related disorders." *Epilepsia*, 2024.
6. **J. Xian**, K.M. Thalwitzer, J. McKee, ..., I. Helbig. "Delineating clinical and developmental outcomes in *STXBP1*-related disorders." *Brain*, 2023.
7. **J. Xian**, I. Helbig. "Guardians of the epilepsy genome." *European Journal of Paediatric Neurology*, 2023.
8. D. deCampo*, **J. Xian***, A. Karlin*, K.R. Sullivan, S.M. Ruggiero, P. Galer, M. Ramos, N.S. Abend, A. Gonzalez, I. Helbig. "Investigating the genetic contribution in febrile infection-related epilepsy syndrome and refractory status epilepticus." *Frontiers in Neurology*, 2023.

9. S.M. Ruggiero, **J. Xian**, I. Helbig. "The current landscape of epilepsy genetics: where are we, and where are we going?" *Current Opinion in Neurology*, 2023.
10. K.M. Thalwitzer, J.H. Driedger, **J. Xian**, ..., S. Syrbe. "Natural history and developmental trajectories of individuals with disease-causing variants in *STXBP1*." *Neurology*, 2023.
11. L. Montanucci, D. Lewis-Smith, R.L. Collins, L.M. Niestroj, S. Parthasarathy, **J. Xian**, ..., D. Lal. "Genome-wide identification and phenotypic characterization of seizure-associated copy number variations in 741,075 individuals." *Nature Communications*, 2023.
12. A. Hanin, J. Cespedes, Y. Pulluru, ..., **J. Xian**, N. Gaspard, L.J. Hirsch. "Review and standard operating procedures for collection of biospecimens and analysis of biomarkers in new onset refractory status epilepticus." *Epilepsia*, 2023.
13. K.R. Sullivan, S.M. Ruggiero, **J. Xian**, ..., I. Helbig. "A disease concept model for *STXBP1*-related disorders." *Epilepsia Open*, 2023.
14. **J. Xian**, S. Parthasarathy, S.M. Ruggiero, ..., I. Helbig. "Assessing the landscape of *STXBP1*-related disorders in 534 individuals." *Brain*, 2022.
15. D. Lewis-Smith, S. Parthasarathy, **J. Xian**, M.C. Kaufman, S. Ganesan, P.D. Galer, R.H. Thomas, I. Helbig. "Computational analysis of neurodevelopmental phenotypes: harmonization empowers clinical discovery." *Human Mutation*, 2022.
16. G. Balagura, **J. Xian**, A. Riva, ..., P. Striano. "Epilepsy course and developmental trajectories in *STXBP1*-DEE." *Neurology: Genetics*, 2022.
17. K. Crawford*, **J. Xian***, K.L. Helbig, ..., I. Helbig. "Computational analysis of 10,860 phenotypic annotations in individuals with *SCN2A*-related disorders." *Genetics in Medicine*, 2021.
18. S. Köhler, M. Gargano, N. Matentzoglu, ..., **J. Xian**, ..., P.N. Robinson. "The Human Phenotype Ontology in 2021." *Nucleic Acids Research*, 2021.
19. S.C. Rametta, S.E. Fridinger, A.K. Gonzalez, **J. Xian**, ..., I. Helbig. "Analyzing 2,589 child neurology telehealth encounters necessitated by the COVID-19 pandemic." *Neurology*, 2020.
20. I. Helbig, G. Barcia, M. Pendziwiat, S. Ganesan, S.H. Mueller, K.L. Helbig, P. Vaidiswaran, **J. Xian**, ..., R. Nabbout, FIRES Genetics Study Group. "Whole-exome and HLA sequencing in febrile infection-related epilepsy syndrome." *Annals of Clinical and Translational Neurology*, 2020.

*** Co-first authors**

PRESENTATIONS & ABSTRACTS

1. **Xian J**, et al. Epigenetic biomarkers of brain injury in children on extracorporeal membrane oxygenation (ECMO). *Crit Care Med.* 2026;54(3S). Abstract 1189. doi:10.1097/01.ccm.0001186752.52641.db
2. **Xian J**, et al. Assessing the clinical and treatment landscape of genetic epilepsies through 3,760 individuals. *Neurology.* 2024;102(17 Suppl 1). Abstract S24.007. doi:10.1212/WNL.0000000000208202
3. Lewis-Smith D, **Xian J**, et al. Expansion of the set of seizure-associated copy number variants to 25 and their clinical characterization in 562 carriers. *Epilepsia.* 2023;64:540.
4. McKee J, **Xian J**, et al. Reconstructing the longitudinal history of *SYNGAP1*-related disorders through data integration across healthcare resources. *Ann Neurol.* 2023;94:S106–S107.

Platforms and Scientific Presentations

1. Serial spatial transcriptomic analyses capture cellular niches in neoadjuvant immune checkpoint inhibition in resectable gastroesophageal cancer.
   UAD Clinical and Translational Meeting. Baltimore, MD, **September 2026**. Oral Presentation.
2. Epigenetic biomarkers of brain injury in critically ill children on extracorporeal membrane oxygenation (ECMO).
   a. Society of Critical Care Medicine 2026 Critical Care Congress. Chicago, IL, **March 2026**. Oral Presentation.
   b. Johns Hopkins Anesthesiology and Critical Care Medicine. Baltimore, MD, **December 2025**. Oral Presentation.
3. Assessing the clinical and treatment landscape of genetic epilepsies through 3,760 individuals. Oral Presentation. American Academy of Neurology Annual Meeting. Denver, CO, **April 2024**.
4. Reconstructing the natural history of *SYNGAP1*-related disorders through data integration. Invited Speaker. 2023 *SYNGAP1* Scientific Conference. Orlando, FL, **November 2023**.
5. Virtual clinical trials and seizure prediction in *STXBP1*-related disorders. Invited Speaker. First Annual European *STXBP1* Summit and Research Roundtable. Milan, Italy, **May 2023**.
6. Longitudinal phenotypes in genetic epilepsies: Reconstructing natural histories from electronic medical records. Invited Speaker. Translating Advancements in Research to Genetic Epilepsy Therapy (TARGETing) Epilepsy Conference, St. Jude Children's Research Hospital. Memphis, TN, **November 2022**.
7. A natural history study of *STXBP1*-related disorders: 259,265 phenotypic footprints reconstructed from medical records. Invited Speaker. *STXBP1* 2022 Annual Investigator's Meeting. Philadelphia, PA, **August 2022**.
8. Leveraging computational approaches to assess neurodevelopmental disorders. Selected Research Talk. Communication Within the Curriculum, University of Pennsylvania. Philadelphia, PA, **April 2022**.
9. Assessing the landscape of *STXBP1*-related disorders. Invited Platform Presentation. American Epilepsy Society Annual Meeting. Chicago, IL, **December 2021**.

Poster Presentations (First Author)

1. **Xian J**, Galer PD, Kaufman MC, Sullivan KR, Ramos M, Gonzalez AK, Back A, Bailey-Medley L, Cohen S, Cuddapah V, Ellis C, Ginn N, Lewin N, Lusk L, Marsh E, Massey S, Ortiz-Gonzalez X, McDonnell P, Ruggiero SM, Taub K, Tefft S, Helbig KL, Goldberg E, Helbig I, Fitzgerald MP. Assessing the clinical and treatment landscape of genetic epilepsies through 3,760 individuals. Oral Presentation. American Academy of Neurology Annual Meeting. Denver, CO, April 2024.
2. **Xian J**, Thalwitzer KM, McKee J, Sullivan KR, Fitch E, Toib J, Kaufman MC, deCampo D, Cunningham K, Brimble E, Fitter N, Syrbe S, Goss J, Rigby C, Pierce S, Ruggiero SM, Prosser P, Boland M, Helbig I. Addressing epilepsy variability through seizure frequency forecasting and virtual clinical trials in *STXBP1*-related disorders. American Epilepsy Society Annual Meeting. Orlando, FL, December 2023.
3. **Xian J**, McKee J, Galer PD, Kaufman M, Helbig I. Forecasting longitudinal seizure trajectories through a novel computational framework based on phenotypic resemblance. International Conference on Artificial Intelligence in Epilepsy and Neurological Disorders. Breckenridge, CO, March 2023.
4. **Xian J**, Ruggiero SM, Fitch E, Cosico M, Thalwitzer KM, Sullivan KR, Kaufman MC, McKee J, Brimble E, Fitter N, Helbig I. A natural history study of *STXBP1*-related disorders using reconstructed medical records. American Epilepsy Society Annual Meeting. Nashville, TN, December 2022.
5. **Xian J**, Parthasarathy S, McKeown S, Balagura G, Fitch E, Helbig K, Gan J, Ellis CA, Lewis-Smith D, Galer PD, Cunningham K, O'Brien M, Baker K, Darling A, Veiga de Goes F, El Achkar CM, Doering JH, Furia F, García-Cazorla A, Gardella E, Geertjens L, Klein C, Kolesnik-Taylor A, Lammertse H, Lee J, Mackie A, Misra-Isrie M,

Olson H, Sexton E, Sheidley B, Smith L, Sotero L, Stamberger H, Syrbe S, Thalwitzer KM, van Berkel A, van Haelst M, Yuskaitis C, Weckhuysen S, Prosser B, Rigby CS, Demarest S, Pierce S, Zhang Y, Møller RS, Bruining H, Poduri A, Zara F, Verhage M, Striano P, Helbig I. Assessing the landscape of *STXBP1*-related disorders. American Epilepsy Society Annual Meeting. Chicago, IL, December 2021.

6. **Xian J**, McKeown S, McKee J, Helbig I. A data-driven analysis of changes in anti-seizure medications after genetic diagnosis in 1,598 individuals with genetic epilepsies. MidAtlantic Bioinformatics Conference. Philadelphia, PA, November 2021.

LEADERSHIP & MENTORSHIP

Johns Hopkins MSTP Interview Committee

American Physician Scientist Association (APSA) Undergraduate & Post Baccalaureate Mentorship Program

Undergraduate students

Mayling Chen, JHU BME — Jan 2025–present
Awarded Johns Hopkins University Provost's Undergraduate Research Award (PURA) Fellowship

HOBBIES & INTERESTS

Interests: cycling, running, rock climbing, soccer, tennis, hiking, skiing, scuba diving